

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
MACHINE LEARNING [CSE_4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the traditional K-Means clustering (partitional) algorithm with Euclidean distance to the following 2D dataset: (5)

A)

Point	Coordinates (x, y)
P1	(2, 2)
P2	(3, 3)
P3	(4, 4)
P4	(7, 5)
P5	(8, 7)
P6	(9, 8)

With $k=2$, use P2 and P5 as the initial cluster centers.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Analyze the major differences between agglomerative and divisive hierarchical clustering with respect to: (5)

- Initial step and stopping condition,
- Computational complexity, and Interpretability of the dendrogram.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) You are given data collected from a smart greenhouse monitoring system. The goal is to predict the plant health status (Healthy / Diseased) based on environmental and leaf condition features. (5)

A)

Plant ID	Leafspots	Humidity	Soil pH High	Status
G1	Yes	High	No	Diseased
G2	No	High	No	Healthy
G3	Yes	Low	Yes	Diseased
G4	No	Low	No	Healthy
G5	Yes	High	Yes	Diseased

Apply the ID3 decision-tree algorithm with Entropy and Information Gain to construct a univariate tree (depth = 1) that predicts the *Status* of each plant.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Examine the relationship between pre-pruning, the bias-variance trade-off, and the resulting generalization performance of decision tree models. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Describe the reason that causes the overall entropy after a dataset split to remain greater than zero when one child node is perfectly pure (entropy = 0.0) and the other has entropy 0.97, both being of equal size. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 3) Examine the role of posterior cluster probabilities in quantifying ambiguity in data point assignments and analyze the limitations of deterministic methods such as K-Means in representing uncertainty within overlapping or noisy datasets. (4)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain how Occam's Razor, the Ugly Duckling Theorem, and the No Free Lunch Theorem together highlight the challenges in building generalizable machine learning models. Support your answer with one example scenario where these principles come into conflict. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) A random forest model shows higher test accuracy than a single decision tree. Analyze what property of bagging contributes to this improvement. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- 4) Differentiate the mathematical structure of the splitting criterion employed in a univariate decision tree from that of a multivariate decision tree. Determine how the reliance on a single feature axis in the univariate approach enforces axis-aligned splits. (4)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply the SVM margin (hard margin) formula to determine the margin of a linear classifier defined by (3)
- $$4x_1 - 3x_2 + 6 = 0.$$
- If the two closest data points (support vectors) lie on the parallel lines
- $$4x_1 - 3x_2 + 12 = 0 \text{ and } 4x_1 - 3x_2 = 0,$$
- Solve the margin of this classifier.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Analyze how the choice of combination rule (majority, weighted, or median voting) impacts ensemble robustness against noisy base-learners. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- 5) (5)
- A)

Apply the following linear regression model to predict crop yield (in kg) based on fertilizer (x_1 , in kg) and rainfall (x_2 , in mm):

$$\hat{y} = 50 + 3x_1 + 0.2x_2$$

1. Compute the predicted yield when fertilizer is 10 kg and rainfall is 100 mm.
2. If fertilizer is increased by 5 kg and rainfall increases by 20 mm, calculate the new predicted yield.

Interpret which factor (fertilizer or rainfall) has a greater influence on yield and justify your reasoning.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Differentiate the roles of the ϵ -insensitive zone and the regularization parameter C in Support Vector Regression (SVR). Determine how increasing the value of C structurally affects the tolerance for training errors and the resulting width of the margin. Finally, analyze the precise mechanism by which a higher C value shifts the model's behaviour towards reduced bias but concurrently increases its variance and sensitivity to outlying data points. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4032:MACHINE LEARNING

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	5
3	3	CO 3	8
4	4	CO 4	14
5	5	CO 5	14
6	6	CO 6	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4032:MACHINE LEARNING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	5(10%)
3	Application	4	18(36%)
4	Analysis	7	27(54%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)